Research Methods and Dissertation

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Research Methods and Dissertation (A32367)

Short Title:	Research Research		
Faculty	Science and Computing		
Department	Science		
Cluster	Placement and Projects		
Author	JGERAGHTY		
Credits	30	**Level:**	Postgraduate

Description of Module / Aims

This module will develop the learner's ability to access relevant scientific information from primary literature, collate the information, and write an overview of a chosen research topic. Participants will learn modern techniques of data analysis in the context of analytical method development and how to apply statistical analysis processes to data sets and interpret the results in order to quantify data quality and compare data sets. Projects undertaken will be linked to a relevant sectoral issue, be it agricultural production, crop and livestock management practices, new technologies, food processing, marketing produce or direct consumer retail. Projects related to this research topic will be carried out under the supervision of individual members of the academic staff and, where relevant, a co-supervisor from organic or biological agriculture.

Programmes

`RESA-0210`	MSc in Organic and Biological Agriculture (`WD_SOBIA_R`)	2 / 4 / M

Indicative Content

- Literature search strategies and the use of online databases to research the scientific literature
- Preparation and critical evaluation of literature for a review related to a scientific research topic
- Technical writing skills, writing science correctly & designing report templates
- Planning, design and implementation of a scientific project in the context of a literature review
- Precision and error measurement: relative and fixed bias, random errors, propagation of errors and accuracy, comparing and combining standard deviations and means of data sets, limit of detection
- Sampling distributions: point estimation, sampling from continuous and discrete distributions
- Statistical inference: inference for single, two or more samples, including t-test, ANOVA, correlations, linear regression, and principal component analysis
- Implementation and interpretation of generalized linear models(GLMs)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Locate and review information from the primary literature relating to a given research topic and determine its relative importance.
2. Construct and develop a cohesive and fully referenced literature review report on a given research topic that conforms to a high standard of technical writing.
3. Analyse the parameters and components of an area of study, presenting these in the form of a hypothesis or problem statement and describing a methodology for investigation.
4. Assess and interpret data sets based on applied inter-comparison significance tests.
5. Synthesise and prepare analytical data using statistical software packages for scientific papers and reports.
6. Develop and demonstrate powers of analysis in the exploration and resolution of research problems.
7. Critique, evaluate and summarize key findings generated on research problems.

Learning and Teaching Methods

- Online lectures and webinars
- Tutorials and supervisor review sessions
- Dissertation production supervision

Assessment Methods

	Weighing	Outcomes Assessed
Dissertation	60%	1,2,3,4,5,6,7
Continuous Assessment	40%	1,3,4,5,6
In-Class Assessment	20%	1,3,4,6
Presentation	20%	3,5,6

Assessment Criteria

<40%: Insufficient level of research work performed in terms of quality and quantity of research output and competence with regard to data analysis skills. Inadequate knowledge on research topic evident through poor report writing and oral presentation skills.

40%--59%: The learner will be able to demonstrate a good understanding of the different aspects to soil health and how these can be assessed and measured. The student will be able to conduct and write a case study/resource audit.

60%--69%: Addressing and analysing the main points and showing evidence of underpinning knowledge. Evidence of evaluation, data analysis skills, and synthesis of the relevant issues and theory to practice integration.

70%--100%: Good level of critical data and information analysis, originality of thought and a comprehensive knowledge base. Good critical evaluation and synthesis of relevant issues, showing an ability to integrate theory and practice.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial		24
Online Delivery		24
Independent Learning		762

Essential Material

- "Academic Search Complete - WIT library databases." https://library.wit.ie/databases/
- "Science Direct database - WIT library databases." https://library.wit.ie/databases/
- "Web of Science database - WIT library databases." https://library.wit.ie/databases/

Supplementary Material

- Carter , M. _Designing Science Presentations_. Oxford, UK: Elsevier Ltd., 2013.
- Glaz, B. and K.M. Yeater. _Applied Statistics in Agricultural, Biological and Environmental Sciences_. USA: John Wiley & Sons, 2020.
- Lebrun, J.L. _Scientific Writing 2.0: A reader's and writer's guide._. New Jersey, USA: World Scientific Publishing Co., 2011.

- Marder, M.P. _Research Methods for Science_. UK: Cambridge University Press, 2011.

Requested Resources

- Room Type: Computer Lab